

Exercise-7

Problem Statement: Financial Forecasting

Source code:

```
using System;
class Program
{
    static double FutureValue(double presentValue, double growthRate, int years)
    {
        if (years == 0)
            return presentValue;
        return FutureValue(presentValue, growthRate, years - 1) * (1 + growthRate /
100);
    }
    static void Main()
    {
        Console.Write("Enter present value: ");
        double presentValue = Convert.ToDouble(Console.ReadLine());
        Console.Write("Enter annual growth rate (%): ");
        double growthRate = Convert.ToDouble(Console.ReadLine());
        Console.Write("Enter number of years: ");
        int years = Convert.ToInt32(Console.ReadLine());
        double futureValue = FutureValue(presentValue, growthRate, years);
        Console.WriteLine("\nFuture Value: " + futureValue);
        Console.WriteLine("\nTime Complexity: O(n)");
        Console.WriteLine("Space Complexity: O(n)");
        Console.WriteLine("Optimization: Use iteration or memoization to avoid
excessive recursive calls.");
    }
}
```

```
}  
}
```

Output:

```
Output  
▲ Enter present value: 1000  
Enter annual growth rate (%): 3  
Enter number of years: 4  
  
Future Value: 1125.50881  
  
Time Complexity: O(n)  
Space Complexity: O(n)  
Optimization: Use iteration or memoization to avoid excessive recursive calls.
```